

Assignment 2

AIML CLZC416 — Mathematical Foundations for Machine Learning

Instructions

1. This is not a group activity. Each student should do the problems and submit individually.
 2. Assignments should be uploaded as single pdf with name BITSID.pdf.
 3. Submissions beyond the deadline would not be graded.
 4. Assignments sent via email would not be accepted.
 5. Copying is strictly prohibited. Adoption of unfair means would lead to disciplinary action.
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Q1 Consider the quadratic objective function (1+1+1 marks)

$$J(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix} \mathbf{w} - [1 \quad 2] \mathbf{w},$$

where

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}.$$

Initial approximation is

$$\mathbf{w}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Write a Python program to implement the Steepest Descent Method with Exact Line Search without using optimization libraries such as `scipy.optimize`.

Your program should:

1. Compute the gradient
2. Compute the steepest descent direction

$$\mathbf{g}_t = -\nabla J(\mathbf{w}_t).$$

3. Determine the optimal step size

$$\alpha_t = \arg \min_{\alpha} J(\mathbf{w}_t + \alpha \mathbf{g}_t)$$

4. Update the parameter vector using

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha_t \mathbf{g}_t.$$

5. Continue the iterations until

$$\|\nabla J(\mathbf{w})\| < 10^{-6}.$$

6. Plot

- Objective function versus iteration.
- Gradient norm versus iteration.

7. Verify numerically that

$$\mathbf{g}_t^T \nabla J(\mathbf{w}_{t+1}) \approx 0.$$

8. Report the optimal solution and the total number of iterations.

Deliverables: Code, plots, the optimal solution and the total number of iterations.

Q2 Consider the quadratic objective function

(0.5+1+0.5+1 marks)

$$g(w) = (w - 1.5)^2, \quad w \in [1, 2]$$

1. Show the Gradient Descent iteration for this function with a learning rate of 0.2 and initial point $w_0 = 2$.

2. Perform 10 iterations of gradient descent (using a Python program) and plot results to show convergence.

Deliverables: Tabulate $w_k, g(w_k)$ for $k = 0, 1, \dots, 10$ for gradient descent and corresponding plot of w vs $g(w)$ at different iterates.

3. Show the Adam iteration for this function, including the bias-corrected learning rate α_t , with a learning rate of 0.2., decay parameters $\rho = 0.999, \rho_f = 0.9$ and initial point $w_0 = 2$.

4. Perform 10 iterations of Adam (using a Python program) and plot results to show convergence.

Deliverables: Tabulate $w_k, A_k, g(w_k)$ for $k = 0, 1, \dots, 10$ for gradient descent and corresponding plot of w vs $g(w)$ at different iterates.

Q3 Consider the function $f(\mathbf{x}) = x_1^4 + 16x_2^4 - 8x_1x_2$

(0.5+1.5+1.5+0.5 marks)

1. Considering the curvature of f along x_1 and x_2 direction, comment whether the gradient descent method with a constant step size is good approach with justification.

2. Starting from the initial point $(1, 1)$, perform 20 iterations of the AdaGrad method using a fixed step size $\alpha = 0.01$ and $\epsilon = 0$.

Deliverables: Tabulate the iterates $x_1^{(k)}, x_2^{(k)}, f_{x_1}^{(k)}, f_{x_2}^{(k)}, A_{x_1}^{(k)}, A_{x_2}^{(k)}$ $k = 1, \dots, 20$ with accuracy up to 6 places after decimal.

3. Starting from the initial point $(1, 1)$, perform 20 iterations of the RMSProp method using a fixed step size $\alpha = 0.01, \rho = 0.9$ and $\epsilon = 0$.

Deliverables: Tabulate the iterates $x_1^{(k)}, x_2^{(k)}, f_{x_1}^{(k)}, f_{x_2}^{(k)}, A_{x_1}^{(k)}, A_{x_2}^{(k)}$ $k = 1, \dots, 20$ with accuracy up to 6 places after decimal.

4. From the results obtained in d) and e), determine whether AdaGrad or RMSProp performs better. Support your conclusion with appropriate reasoning. Mere identification of the better optimizer without justification will receive 0 marks.